2025-spring Introduction to Quantum Information and Quantum Computing

HW # 4 (Due 6/10 midnight = 6/11 00:00 am). Hand in as a single pdf file through ETL. For Qiskit problems: Pasting only the result (output) of the code (perhaps as a picture) will suffice—no need to hand in the coding itself unless explicitly specified. Also, running on an actual backend is recommended if possible, but the simulation result is also Okay.

1. Show that the dynamical map degined in Eqn. 6.69 (lecture note ver 0.85) is a valid quantum map. That is, show that it can be expressed in Kraus form.

2. Show that the Master equation in the Lindblad form discussed in the lecture preserves the trace of the density matrix (internet search is allowed).

3. Bit flip error correction

   A. Consider the three-qubit code for correcting bit flips, where a qubit in the state $|\psi\rangle = a|0\rangle + b|1\rangle$ is encoded as $|\psi_C\rangle = a|000\rangle + b|111\rangle$. Suppose the encoded state is distorted by a rotation about x-axis of the Bloch sphere of the third qubit by amount of 60°. Use the error correction circuit discussed in class and show that the state $|\psi_C\rangle$ is recovered after error correction.

   B. We know that any single qubit quantum process can be expressed in the form $\sum_i E_i \rho E_i^\dagger$, with $E_i = e_{i0} I + e_{ix}\sigma_x + e_{iy}\sigma_y + e_{iz}\sigma_z$. Give the discretization of the error process explicitly for this 60°rotation.

4. **Textbook exercise. 8.2, 8.3** You can of course refer to the solution provided in the textbook. However, I trust that you do not just copy and paste it.

Exercise 8.2 Consider a function $\mathcal{E}_t$ that transforms the initial state of a qubit ρ into

$$\rho(t) = \mathcal{E}_t(\rho) = f(t)\rho + (1 - f(t))\frac{\mathbb{1}}{2}.$$

Note the similarity between this function and the depolarizing channel presented in Section 7.2. Find the analytical expression of $f(t)$ such that $\mathcal{E}_t$ satisfies the semigroup property $\mathcal{E}_t(\mathcal{E}_s(\rho)) = \mathcal{E}_{t+s}(\rho)$.

Exercise 8.3 In Exercise 8.2, we showed that the quantum map $\mathcal{E}_t(\rho) = e^{-\gamma t}\rho + (1 - e^{-\gamma t})\frac{\mathbb{1}}{2}$ satisfies the semigroup property. Using the identity $\mathbb{1}/2 = \frac{1}{4}(\sigma_x\rho\sigma_x + \sigma_y\rho\sigma_y + \sigma_z\rho\sigma_z + \rho)$, the quantum map $\mathcal{E}_t$ can be written as

$$\mathcal{E}_t(\rho) = \frac{3e^{-\gamma t} + 1}{4}\rho + \frac{1 - e^{-\gamma t}}{4}(\sigma_x\rho\sigma_x + \sigma_y\rho\sigma_y + \sigma_z\rho\sigma_z).$$

Derive the Markovian master equation associated with $\mathcal{E}_t$. **Hint:** Use eqn (8.8) and write $\mathcal{E}_t$ in the form $\mathcal{E}_t(\rho) = \sum_{i=1}^4 \sum_{j=1}^4 c_{ij} F_i \rho F_j^\dagger$ where F_i are the elements of the Pauli basis.

5. Do Exercises of the Qiskit file "Quantum simulation with Trotterization.ipynb" Paste the result

6. Do Exercises of the Qiskit file "Variational Quantum Algorithms.ipynb" Paste the result

7. **Qutip exercise:** Qutip file for this problem is posted on Week 14 module of the ETL. Downlaod and open "NVcenter simulation1.ipynb".

   A. The provided file demonstrates spin qubit Rabi oscillation of the diamond Nitrogen-Vacancy (NV) color center. FYI, this is three spin system consisting of one spin-1 electron spin (Triplet states), one spin-1 Nitrogen nuclear spin, and one spin-1/2 carbon13 spin. Examine the code and put annotations for each line explaining the syntax and meaning of the codings. Paste your annotations. Especially try to express in handwriting how Hamiltonian looks like. You may search internet for basic information about spin Hamiltonian of typical NV center experiments.

   B. The code demonstrates Rabi oscillation simulation with electron and nuclear initial states set as logical basis states. Modify the code so that we observe Ramsey fringe in 2d (Prob. Vs. time and detuning). Also modify initial state so that you can observe coherent oscillations between several different states.